



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 5 Name: Dhanish Ladwani PRN: 24070521020

1. Experiment Title:

Apply array methods and object handling; create a cart total calculator with discount logic.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Experiment 5.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shopping Cart Calculator</title>
  <link rel="stylesheet" href="Experiment 5.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
  <div class="container">
    <h2>Shopping Cart Calculator</h2>

    <div class="input-group">
      <input type="text" id="name" placeholder="Product Name">
      <input type="number" id="price" placeholder="Price">
      <input type="number" id="qty" placeholder="Quantity">
      <button onclick="addProduct()">Add Product</button>
    </div>

    <div class="table-wrapper">
      <table id="cartTable">
        <thead>
          <tr>
            <th>ID</th>
            <th>Product</th>
            <th>Price</th>
            <th>Quantity</th>
            <th>Total</th>
          </tr>
        </thead>
        <tbody>
          </tbody>
        </tbody>
      </table>
    </div>

    <div class="result-card">
      <div id="result" class="totals-summary">
        <p><b>Total Amount :</b> 0.00</p>
        <p><b>Discount :</b> 0.00</p>
      </div>
    </div>
  </div>
</body>
</html>
```

```

        <p><b>Final Amount :</b> 0.00</p>
    </div>

    <div class="lists-container">
        <div class="summary-section">
            <h3>Item Summary</h3>
            <ul id="summary"></ul>
        </div>
        <div class="expensive-section">
            <h3>Expensive Products (Price > 1000)</h3>
            <ul id="expensive"></ul>
        </div>
    </div>
</div>

<footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-top: 1px solid lightgray; padding-top: 15px;">
    Made by Dhanish Ladwani PRN - 24070521020
</footer>
</div>

<script src="Experiment 5.js"></script>
</body>
</html>

```

Experiment 5.css

```

* {
    box-sizing: border-box;
}

body {
    background-color: whitesmoke;
    color: dimgray;
    font-family: sans-serif;
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 700px;
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 24px;
    font-size: 24px;
}

.input-group {
    display: flex;
    gap: 12px;
    margin-bottom: 24px;
    flex-wrap: wrap;
}

input {
    flex: 1 1 140px;
}

```

```

    min-width: 0;
    padding: 10px 14px;
    border: 1px solid lightgray;
    border-radius: 6px;
    font-size: 14px;
    outline: none;
    transition: border-color 0.2s;
}

input:focus {
    border-color: royalblue;
}

button {
    padding: 10px 20px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 14px;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s;
}

button:hover {
    background-color: blue;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-bottom: 24px;
    background-color: white;
    border-radius: 6px;
    overflow: hidden;
    border: 1px solid lightgray;
}

th, td {
    padding: 12px;
    border-bottom: 1px solid lightgray;
    text-align: center;
    font-size: 14px;
}

th {
    background-color: royalblue;
    color: white;
    font-weight: 600;
}

tr:nth-child(even) {
    background-color: whitesmoke;
}

tr:last-child td {
    border-bottom: none;
}

.result-card {
    background-color: whitesmoke;
    padding: 20px;
    border: 1px solid lightgray;
    border-radius: 8px;
    line-height: 1.8;
    font-size: 15px;
}

```

```

.totals-summary p {
  margin: 6px 0;
  font-size: 15px;
  color: #333;
}

.totals-summary p:nth-child(2) {
  color: #d9534f;
  font-weight: bold;
}

.totals-summary p:nth-child(3) {
  color: #28a745;
  font-weight: bold;
  font-size: 17px;
}

.lists-container {
  margin-top: 20px;
  border-top: 1px solid lightgray;
  padding-top: 15px;
}

h3 {
  color: darkblue;
  font-size: 16px;
  margin-top: 15px;
  margin-bottom: 8px;
}

ul {
  padding-left: 20px;
  margin-top: 4px;
  margin-bottom: 12px;
  color: dimgray;
}

```

Experiment 5.js

```

let cart = [];

function addProduct() {
  let name = document.getElementById("name").value.trim();
  let price = parseFloat(document.getElementById("price").value);
  let qty = parseInt(document.getElementById("qty").value);

  if (name === "" || isNaN(price) || isNaN(qty) || price <= 0 || qty <= 0) {
    alert("Please enter valid product details");
    return;
  }

  let product = {
    id: cart.length + 1,
    name: name,
    price: price,
    quantity: qty
  };

  cart.push(product);
  displayCart();

  document.getElementById("name").value = "";
  document.getElementById("price").value = "";
}

```

```

    document.getElementById("qty").value = "";
}

function displayCart() {
    let tableBody = document.querySelector("#cartTable tbody");
    if (!tableBody) {
        let table = document.getElementById("cartTable");
        tableBody = document.createElement("tbody");
        table.appendChild(tableBody);
    }

    tableBody.innerHTML = "";

    cart.forEach(function(item) {
        tableBody.innerHTML += `
            <tr>
                <td>${item.id}</td>
                <td>${item.name}</td>
                <td>${item.price.toFixed(2)}</td>
                <td>${item.quantity}</td>
                <td>${(item.price * item.quantity).toFixed(2)}</td>
            </tr>
        `;
    });

    let total = cart.reduce(function(sum, item) {
        return sum + (item.price * item.quantity);
    }, 0);

    let discount = 0;
    let discountPercent = 0;

    if (total >= 50000) {
        discountPercent = 20;
        discount = total * 0.20;
    } else if (total >= 20000) {
        discountPercent = 10;
        discount = total * 0.10;
    } else if (total >= 5000) {
        discountPercent = 5;
        discount = total * 0.05;
    }

    let finalAmount = total - discount;

    document.getElementById("result").innerHTML = `
        <p><b>Total Amount :</b> ₹${total.toFixed(2)}</p>
        <p><b>Discount (${discountPercent}%) :</b> -₹${discount.toFixed(2)}</p>
        <p><b>Final Amount :</b> ₹${finalAmount.toFixed(2)}</p>
    `;

    let summary = document.getElementById("summary");
    summary.innerHTML = "";
    cart.map(function(item) {
        summary.innerHTML += `<li>${item.name} : ${item.price * item.quantity}</li>`;
    });

    let expensive = document.getElementById("expensive");
    expensive.innerHTML = "";

    let exp = cart.filter(function(item) {
        return item.price > 1000;
    });

    exp.forEach(function(item) {
        expensive.innerHTML += `<li>${item.name} (${item.price.toFixed(2)})</li>`;
    });
}

```

4. Output:

The image displays two screenshots of a web application titled "Shopping Cart Calculator". The browser's address bar shows the file path: `C:/Users/Admin/Downloads/Javascript/Experiment%205/experiment%205/Experiment%205.html`.

Top Screenshot:

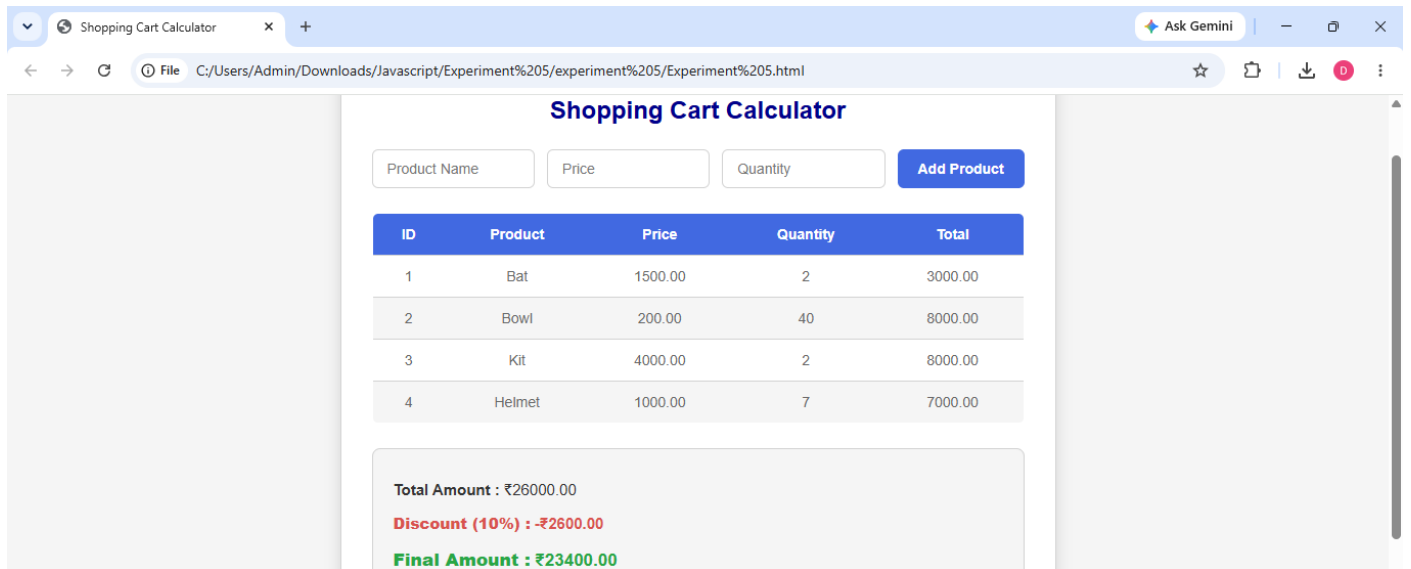
- Form:** Input fields for "Product Name", "Price", and "Quantity", followed by an "Add Product" button.
- Table:** A table with 5 columns: ID, Product, Price, Quantity, and Total. It contains one row: ID 1, Product Bat, Price 1500.00, Quantity 2, Total 3000.00.
- Summary:**
 - Total Amount : ₹3000.00
 - Discount (0%) : -₹0.00
 - Final Amount : ₹3000.00
- Item Summary:**
 - Bat : 3000.00
- Expensive Products (Price > 1000):**
 - Bat (1500.00)

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Bottom Screenshot:

- Form:** Same as the top screenshot.
- Table:** A table with 5 columns: ID, Product, Price, Quantity, and Total. It contains three rows:

ID	Product	Price	Quantity	Total
1	Bat	1500.00	2	3000.00
2	Bowl	200.00	40	8000.00
3	Kit	4000.00	2	8000.00
- Summary:**
 - Total Amount : ₹19000.00
 - Discount (5%) : -₹950.00
 - Final Amount : ₹18050.00
- Item Summary:**
 - Bat : 3000.00
 - Bowl : 8000.00
 - Kit : 8000.00



5. Result/Conclusion:

A shopping cart calculator was developed using arrays of objects to store product details such as name, price and quantity. The `forEach()` method was used to render each cart item into an HTML table, while `reduce()` was used to compute the total cart amount. Conditional logic was applied on the total to calculate tiered discounts, and `map()` and `filter()` were used to generate an item-wise summary and to list products priced above a threshold respectively. This experiment reinforced the use of array methods (`map`, `filter`, `reduce`, `forEach`) and objects for managing and processing structured data in JavaScript.